



2026 BASIC ABSTRACT ALGEBRA STUDY

PRE-COURSE SELF ASSESSMENT

SOLUTION - Q5

Version

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Read this carefully before moving on.

In the following questions, *prove or disprove* means you need to either show a correct logical proof of a given statement or provide at least one counterexample that disproves the statement. Describe each deduction step by step, citing the relevant definition or fact at each step.

As a tiny example of a proof answer, consider the following statement.

For any integers a and b , if a and b are both even, then so is $a + b$.

The above statement is true.

Proof. Assume a and b are even integers. By the definition of an even integer, there exist integers k and l such that $a = 2k$ and $b = 2l$. Then $a + b = 2k + 2l = 2(k + l)$, and since $k + l$ is an integer, $a + b$ is even by the same definition. $\square$

As a tiny example of a disproof answer, consider the following statement.

For any integers a and b , if $a + b$ is even, then a and b are both even.

The above statement is false.

Proof. Take $a = 1$ and $b = 1$. Both are odd, but $a + b = 2$ is even. Hence the statement is false. $\square$

It is highly encouraged to write down full sentences like the above samples, rather than just throwing out a few symbols or numbers without explaining why.

Note 1

The following symbols will be used throughout the following questions.

- $\mathbb{Z}$: A set of all integers.
- $\mathbb{Z}^+$: A set of all positive integers. i.e., $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$.
- $\mathbb{Z}_n$: A set of non-negative integers from 0 to $n - 1$ for some positive integer n . For instance, $\mathbb{Z}_4 = \{0, 1, 2, 3\}$.
- $\mathbb{Q}$: A set of all rational numbers.
- $\mathbb{R}$: A set of all real numbers.
- $\mathbb{R}^+$: A set of all positive real numbers.
- $\mathbb{C}$: A set of all complex numbers.

Definition 2 (Extensionality)

Two sets X and Y are said to be equal if and only if X is a subset of Y and Y is a subset of X . This is called the *axiom of extensionality*. If two sets X and Y are equal, we write $X = Y$.

Q1.

Note 3. $A \setminus B$ denotes the set of all elements in A that are not in B .

Which of the following is distinct from the others?

- ☐ 1. $\{-x \mid x \in \mathbb{R}\}$
☐ 2. $\{xx \mid x \in \mathbb{R}\}$
☐ 3. $\{xy \mid x, y \in \mathbb{R}\}$
- ☐ 4. $\{xxx \mid x \in \mathbb{R}\}$
☐ 5. $\{\frac{x}{y} \mid x \in \mathbb{R}, y \in \mathbb{R} \setminus \{0\}\}$

Q2.

Note 4. The sum of two integers is always an integer.

Show that $\{x + y \mid x, y \in \mathbb{Z}\} = \mathbb{Z}$ using extensionality. (Definition 2)

Hint 5. First, show that $\{x + y \mid x, y \in \mathbb{Z}\} \subset \mathbb{Z}$. Then, show that $\mathbb{Z} \subset \{x + y \mid x, y \in \mathbb{Z}\}$. In the latter case, it is enough to show that for any integer, there exist at least a pair of integers such that their sum gets equal to that integer.

Definition 6 (Size of finite sets)

A size of a finite set X is the number of elements it contains, and denoted as $|X|$.

Q3.

Let A be a set of all prime numbers that are even. What is the value of $|A|$?

- ☐ 1. 0
 ☐ 2. 1
 ☐ 3. 2
 ☐ 4. 3
 ☐ 5. 4

Q4.

Let $A = \{n + m \mid n, m \in \mathbb{Z}_{20261212}\}$. What is the value of $|A|$?

Hint 7. $|\{n + m \mid n, m \in \mathbb{Z}_2\}| = |\{0 + 0, 0 + 1, 1 + 0, 1 + 1\}| = |\{0, 1, 2\}| = 3$.

Answer: _____

Q5.

Let T be the set of all real numbers except 0 and 1. Define two functions $f, g : T \rightarrow T$ as $f(x) = x^{-1}$ and $g(x) = 1 - x$, respectively.

Let S be the set of all possible functions that can be created by composing f and g any number of times in any order, such as $f, g, g \circ f, f \circ g, g \circ g \circ f \circ f, f \circ g \circ f, f^{20261212}$, etc.

What is the value of $|S|$?

Answer: 6

Proof. Let 1 denote the identity function on T , i.e., $1(x) = x$ for all $x \in T$. For convenience, we will write gf to denote the composition $g \circ f$.

Keep in mind that having x or $1 - x$ in the denominator is perfectly valid, as T contains neither 0 nor 1 .

It is easily observed that $f^2 = g^2 = 1$ as $f^2(x) = (x^{-1})^{-1} = x$ and $g^2(x) = 1 - (1 - x) = x$, respectively. It directly follows that in any sequence of compositions, consecutive identical functions cancel out. Odd number of repetitions get shrunk to only one occurrence, while even number of repetitions are completely cancelled out as they are equal to the identity 1 . For instance, $gfffg$ is the same as gfg , and $gfgggg$ is the same as gf . This implies that any unique function in S must be a strictly alternating sequence of f and g , such as $fgfgf$.

Next, let's evaluate the alternating sequence of length six, $gfgfgf$:

$$gfgfgf(x) = gfgf\left(1 - \frac{1}{x}\right) = gf\left(\frac{1}{1-x}\right) = x$$

Since $gfgfgf(x) = x$, we establish that $gfgfgf = 1$.

Based on this identity, we can compose both sides with g or f on the left, moving terms on the left to the right, once at a time. Keeping in mind that $g^2 = 1$ and $f^2 = 1$, we can find all equivalent alternating sequences as follows.

$$\begin{aligned} gfgfgf &= 1 \\ ffgfgf &= g && \text{(multiply by } g \text{ on the left)} \\ gfgfgf &= fg && \text{(multiply by } f \text{ on the left)} \\ fgfgfg &= gfg && \text{(multiply by } g \text{ on the left)} \\ gfgfgf &= fgfg && \text{(multiply by } f \text{ on the left)} \\ fgfgfg &= f && \text{(multiply by } g \text{ on the left)} \\ 1 &= fgfgfg && \text{(multiply by } f \text{ on the left)} \end{aligned}$$

From here, we can ensure that every possible composition of f and g is essentially the same as one of the values above, as any alternating pattern longer than 6 cycles back to the identity. For instance, $fgfgfgfg$ is the same as $(fgfgfg)f = 1f = f$, and so on.

Consequently, any sequence longer than 3 can be rewritten as a shorter equivalent sequence. By keeping only the shorter side of each equation, we collect exactly six candidates.

$$1 \quad f \quad g \quad gf \quad fg \quad fgf$$

Finally, we can explicitly evaluate the remaining compositions to confirm they are all distinct.

$$(gf)(x) = 1 - \frac{1}{x} \quad (fg)(x) = \frac{1}{1-x} \quad (fgf)(x) = \frac{x}{x-1}$$

Since each expression yields a distinct rational function, there are exactly six distinct functions in the set S .

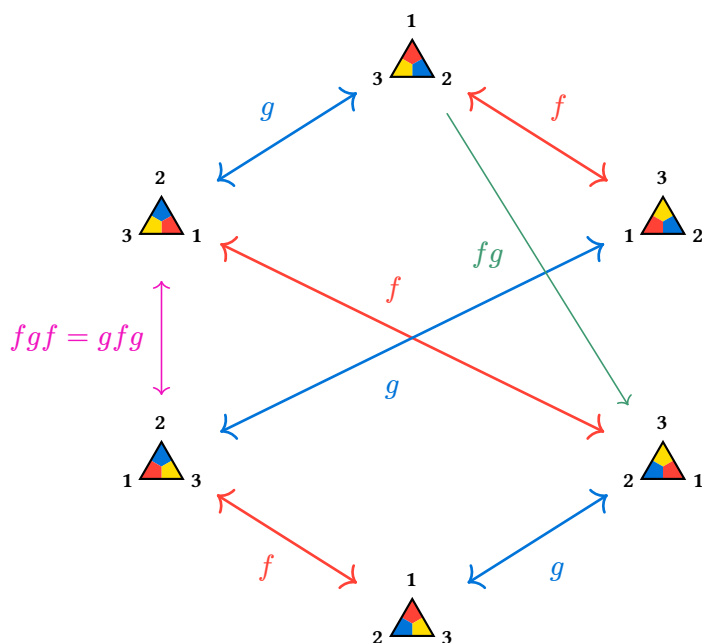
Therefore, the correct answer is 6. □

Because every possible composition of f and g results in a function that is already in S , we say that the set S is *closed* under function composition. Moreover, we can always find a pair of functions that undo each other's action when composed together. Specifically, 1 , f , g , and fgf are self-inverses, whereas fg and gf are inverses of each other. In fact, such a set S forms a *group* under function composition.

One of the most famous names for this set S is the *symmetric group of degree 3*, denoted as S_3 . It represents all possible arrangements of three distinct elements. We say that S_3 is *generated* by f and g . In this context, f and g are called *transpositions* because their action swaps exactly two elements while leaving the third one perfectly in place.

This group S has another famous name, the *dihedral group of degree 3*, denoted as D_3 . This group captures all the rigid *symmetries* of an equilateral triangle. We can visualize f as a reflection (flip) of the triangle across an axis passing through the bottom-right corner and the center, while g acts as a reflection across an axis passing through the bottom-left corner and the center.

The remaining four symmetries can be generated entirely by chaining these two basic reflections. The diagram below is a *Cayley graph* of D_3 , mapping out the entire group. Here, applying f is represented by following a red arrow, and applying g is represented by a blue arrow.



To understand how this works, let's trace a path. Suppose we want to move from the starting triangle positioned at $(1, 2, 3)$ (reading top, bottom-right, bottom-left) to the triangle positioned at $(3, 1, 2)$. We can first follow the blue arrow (g) to $(2, 1, 3)$, and then follow the red arrow (f) to arrive at $(3, 1, 2)$. Because we apply g first and then f , the resulting function composition is written right-to-left as fg . Geometrically, this combined motion is exactly equivalent to rotating the triangle by 120 degrees clockwise.

Similarly, imagine navigating from the triangle at $(2, 3, 1)$ to $(2, 1, 3)$. We could trace the blue-red-blue path upward and then downward (g then f then g , forming gfg). Alternatively, we could trace the red-blue-red path downward and then upward (f then g then f , forming fgf). Both paths lead to the exact same destination, intuitively showing that $gfg = fgf$. If we look at the numbers on the triangles, this specific move swaps the bottom two corners while fixing the top, acting exactly like a reflection across the vertical axis, which makes the arrow bidirectional just like the f and g .

Notice that the red f and blue g arrows are bidirectional. This visually encodes the fact that f and g are self-inverses ($f^2 = 1$ and $g^2 = 1$). Because traversing an arrow back and forth simply cancels out the motion, simplifying an algebraic expression made of f 's and g 's is essentially the same puzzle as finding the shortest, non-backtracking path between two nodes on this graph.

By simply chasing the arrows in our head, we can visually confirm why any sequence of f and g , no matter how long, is completely trapped within these six perfectly closed states.

Q6.

Let $\text{Di}_6 = \{\square, \square, \square, \square, \square, \square\}$ be a set of all possible faces of a six-sided die.

How many distinct ordered pairs can be made from Di_6 ?

- ☐ 1. 36
 ☐ 2. 21
 ☐ 3. 6
 ☐ 4. 30
 ☐ 5. 12

Definition 8 (Product of sets)

Let X, Y be two sets. A *product* $X \times Y$ between them is a set of all ordered pairs (x, y) where $x \in X$ and $y \in Y$.

Q7.

Let $X = \{10, 20, 30\}$ and $Y = \{\leftarrow, \rightarrow\}$. Find the same set as $Y \times X$.

- ☐ 1. $\{(10, 10), (20, 20), (30, 30), (\leftarrow, \leftarrow), (\rightarrow, \rightarrow)\}$
☐ 2. $\{(10, \leftarrow), (10, \rightarrow), (20, \leftarrow), (20, \rightarrow), (30, \leftarrow), (30, \rightarrow)\}$
☐ 3. $\{(\leftarrow, 20), (20, \leftarrow), (\leftarrow, 20), (20, \leftarrow), (30, \rightarrow), (\rightarrow, 30)\}$
☐ 4. $\{(10, \rightarrow), (20, \rightarrow), (30, \rightarrow), (10, \leftarrow), (20, \leftarrow), (30, \leftarrow)\}$
☐ 5. $\{(\leftarrow, 20), (\leftarrow, 30), (\leftarrow, 10), (\leftarrow, 20), (\rightarrow, 10), (\rightarrow, 20), (\rightarrow, 30)\}$

Definition 9 (Power set)

A set of all subsets of a set X is called a *power set* of X , and denoted as $\mathcal{P}(X)$.

Q8.

Which of the following is the same as $\mathcal{P}(\{a, 1, *\})$, if $a, 1$, and $*$ are all distinct?

- ☐ 1. $\{\{\emptyset, \{a\}, \{1\}, \{*\}, \{a, 1\}, \{1, *\}, \{a, *\}, \{a, 1, *\}\}\}$
☐ 2. $\{\{*, 1, a\}, \{*, a\}, \{*, 1\}, \{1, a\}, \{*\}, \{a\}, \{1\}\}$
☐ 3. $\{\emptyset, \{1\}, \{a\}, \{*\}, \{(a, 1)\}, \{(1, *)\}, \{(a, *)\}, \{(1, a, *)\}\}$

- 4. $\{\{*, 1, a\}, \{a\}, \{a, 1, a\}, \{*, *\}, \{1\}, \emptyset, \{1, *\}, \{a\}, \{a, *, *\}\}$
- 5. $\{\emptyset, \{a\}, \{1\}, \{*\}, \{a, 1\}, \{1, a\}, \{1, *\}, \{*, 1\}, \{a, 1, *\}\}$

Q9.

What is the value of $|\mathcal{P}(\mathbb{Z}_2 \times \mathbb{Z}_5)|$?

Answer: _____

Q10.

Prove or disprove that the following holds for any two sets A and B .

$$(A \setminus B) \cap (B \setminus A) = \emptyset$$

Q11.

Prove or disprove that the following holds for any set X .

$$\mathcal{P}(X) \cap X = \emptyset$$

Definition 10 (Singleton)

A set is called a *singleton* if it has only one element.

Q12.

Prove or disprove the following statement.

A set is a singleton if and only if it has only one proper subset.

Definition 11 (Injection, Surjection, and Bijection)

A function $f : X \rightarrow Y$ is said to be *injective* or called an *injection* if for every $x, y \in X$, if $x \neq y$ then $f(x) \neq f(y)$. A function is said to be *surjective* or called a *surjection* if for every $y \in Y$, there exists an $x \in X$ such that $f(x) = y$. A function is said to be *bijective* or called a *bijection* if it is both an injection and a surjection.

Q13.

For a set X , think of another set $X/=$ that is a set of all singletons from all elements of X . For instance, if X is $\{a, b, c\}$, then $X/=$ becomes $\{\{a\}, \{b\}, \{c\}\}$.

Prove or disprove that there is a bijection between X and $X/=$.

Definition 12 (Equality of functions)

Two functions with the same domain and same codomain are said to be *equal* if and only if they map every element of the domain to the same element of the codomain.

In other words, if $f, g : X \rightarrow Y$ are functions with the same domain and codomain, then $f = g$ if and only if $f(x) = g(x)$ for every $x \in X$.

Q14.

Show that for any singleton, there is always a unique function from any set to it.

Hint 13. Let $\{*\}$ be a singleton and X be any set, and consider a function $f : X \rightarrow \{*\}$. First show that such function always exists, and then prove that if there is a function $f' : X \rightarrow \{*\}$, then $f = f'$.

Q15.

Note 14. An induction here refers to the mathematical induction.

Let A_i be a singleton for $i = 1, \dots, n$ for a positive integer n . Show by induction that $A_1 \times \dots \times A_n$ is also a singleton.

Q16.

Let X, Y be finite sets. How many functions $f : X \rightarrow Y$ are there from X to Y ?

- ☐ 1. $|Y|^{|X|}$
☐ 2. $2^{|X| \cdot |Y|}$
☐ 3. $_{|Y|}P_{|X|}$
☐ 4. $|X|^{|Y|}$
☐ 5. $|X|! \cdot |Y|!$

Q17.

Prove or disprove that the following statement holds for any finite sets X, Y and Z .

If the number of bijections from X to Y is the same as the number of bijections from X to Z , then $|Y| = |Z|$.

Q18.

Prove or disprove that if f and g are injective, then $g \circ f$ is injective for any functions f and g .

Q19.

Prove or disprove that if $f \circ g = f \circ h$, then $g = h$ for any functions $f : Y \rightarrow Z$, $g, h : X \rightarrow Y$.

Q20.

Prove or disprove the following statement.

For any finite set X and Y where $|X| = |Y|$, a function $f : X \rightarrow Y$ is injective if and only if it is surjective.

Hint 15. Use the pigeonhole principle.

Q21.

Note 16. $\ln(x)$ denotes the natural logarithm of x .

Prove or disprove that $\ln : \mathbb{R}^+ \rightarrow \mathbb{R}$ is bijective.

Q22.

Which of the following is the same as $(-i)^{4443}$?

- ☐ **1.** $(-i)^{666}$
☐ **2.** i^{321}
☐ **3.** i^{123}
☐ **4.** $(-i)^{4812}$
☐ **5.** $(-i)^{333}$

Q23.

How many trailing zeros are there in the decimal representation of $629!$?

Hint 17. There are two trailing zeros in $11!$, as $11! = 39916800$.

Answer: _____

Q24.

Note 18. For integers a and b , $a \perp b$ denotes that there are no common divisors of a and b other than 1. It is equivalent to saying that $\gcd(a, b) = 1$. In contrast, $a \not\perp b$ denotes that there are common divisors other than 1 between a and b .

- (a) Prove or disprove that the following statement holds for any integers a , b and c .

If $a \perp b$ and $b \perp c$, then $a \perp c$.

- (b) Prove or disprove that the following statement holds for any integers a , b and c .

If $a \not\perp b$ and $b \not\perp c$, then $a \not\perp c$.

Q25.

- (a) Prove or disprove that the following statement holds for any integers a , b and c .

If a divides b and b divides c , then a divides c .

- (b) Prove or disprove that the following statement holds for any integers a , b and c .

If a does not divide b and b does not divide c , then a does not divide c .

Lemma 19 (Euclid's lemma)

For integers a , b , c , if $a \perp b$ and a divides bc , then a divides c .

Q26.

Note 20. For a rational number r , an irreducible form of r is a representation $r = \frac{p}{q}$ such that $p \in \mathbb{Z}$, $q \in \mathbb{Z}^+$, and $p \perp q$.

Define $f : \mathbb{Q} \rightarrow \mathbb{Z} \times \mathbb{Z}^+$ to be the rule that sends each $r \in \mathbb{Q}$ to (p, q) , where $\frac{p}{q}$ is an irreducible form of r .

- (a) Show that f is a well-defined function.

Hint 21. First, show that every rational number admits an irreducible form. Then, show that if $\frac{p}{q}$ and $\frac{p'}{q'}$ are both irreducible forms of the same rational number, then $p = p'$ and $q = q'$.

- (b) Prove or disprove that f is injective.

- (c) Prove or disprove that f is surjective.

Q27.

Show by induction that an equation of the following form has a unique complex solution for each $n \in \mathbb{Z}^+$.

$$x^n = 0$$

Hint 22. If $ab = 0$, then either a or b is zero.

Q28.

Let $\mathbf{X}$ be a $m \times m$ matrix with complex entries and $\mathbf{0}$ be the zero matrix. Prove or disprove that an equation of the following form has a unique solution for each $n, m \in \mathbb{Z}^+$.

$$\mathbf{X}^n = \mathbf{0}$$

Q29.

Let $\mathbf{A}$ be a $n \times n$ matrix with complex entries. Prove or disprove that the following equation has a unique solution for each $n \in \mathbb{Z}^+$.

$$\mathbf{A} = -\mathbf{A}$$

Q30.

Prove or disprove that for any positive integer $n \in \mathbb{Z}^+$, an equation of the following form has at most two distinct solutions in $\mathbb{Z}_n$.

$$x^2 \equiv 1 \pmod{n}$$

Hint 23. By ‘at most two distinct’ solutions, we mean that the number of solutions can be zero, one, or two, but not more than two.

Q31.

Given any $a \in \mathbb{Z}_n$ such that $a \perp n$, prove or disprove that an equation of the following form has a unique solution in $\mathbb{Z}_n$ for each $n \in \mathbb{Z}^+$.

$$ax \equiv 1 \pmod{n}$$

Hint 24. See **Q20**. Can you find a relevant bijection on $\mathbb{Z}_n$?

Q32.

(a) How many solutions does the equation below have in $\mathbb{Z}_{12121212}$?

$$1212x \equiv 4848 \pmod{12121212}$$

Hint 25. Try considering similar questions on $\mathbb{Z}_3$, $\mathbb{Z}_8$, or any small $\mathbb{Z}_n$ and solving them. Can you find a pattern?

Answer: _____

(b) How many solutions does the equation below have in $\mathbb{Z}_{20261212}$?

$$1212x \equiv 2026 \pmod{20261212}$$

Answer: _____

(c) How many solutions does the equation below have in $\mathbb{Z}_{20262121}$?

$$1212x \equiv 2026 \pmod{20262121}$$

Hint 26. See [Q31](#).

Answer: _____

Q33.

Let $\mathbb{Z}^{\mathbb{Z}}$ be a set of all functions from $\mathbb{Z}$ to $\mathbb{Z}$ and define a function $f \in \mathbb{Z}^{\mathbb{Z}}$ by $f(x) = 2x$. We denote the identity function on $\mathbb{Z}$ by $1_{\mathbb{Z}}$.

Prove or disprove that the following equation has a unique solution in $\mathbb{Z}^{\mathbb{Z}}$. In other words, there is a unique function $x : \mathbb{Z} \rightarrow \mathbb{Z}$ such that the following equation holds.

$$x \circ f = 1_{\mathbb{Z}}$$

Q34.

Let Γ be an arbitrary finite set of invertible $n \times n$ matrices with complex entries with $n \in \mathbb{Z}^+$. Suppose that Γ is closed under matrix multiplication. In other words, for any $A, B \in \Gamma$, the product AB is also in Γ .

Let Ψ be the sum of all matrices in Γ .

$$\Psi = \sum_{\Omega \in \Gamma} \Omega$$

Show that for any matrix $\Lambda \in \Gamma$, the equation $\Lambda\Psi = \Psi$ holds.

Hint 27. See [Q20](#).

End.

You may now submit your work.

Evaluation sheet

For each question so far, evaluate it based on the following criteria:

- If you encountered new concepts or new words for the first time, mark ✓ in the New concepts column.
- If you failed to understand the question, mark × in the Evaluation column.
- If you understood the question, but didn't know how to solve it, mark ? in the Evaluation column.
- If you understood the question and are sure you have solved it, mark ○ in the Evaluation column.

No.	New concepts	Evaluation	No.	New concepts	Evaluation
Q1			Q18		
Q2			Q19		
Q3			Q20		
Q4			Q21		
Q5			Q22		
Q6			Q23		
Q7			Q24		
Q8			Q25		
Q9			Q26		
Q10			Q27		
Q11			Q28		
Q12			Q29		
Q13			Q30		
Q14			Q31		
Q15			Q32		
Q16			Q33		
Q17			Q34		

Please answer these additional questions:

- (a) Which was the hardest for you, among those you have solved? _____
- (b) Which seemed to be the hardest for you, among those you haven't solved? _____
- (c) Which was your favorite question, if you had to choose one? _____